

Relaxing the Molecular Clock Assumption: Proof Sketch

1 Shift in Assumptions

To relax the ultrametric tree (molecular clock) assumption, we transition from a strictly constant block model to a bounded block model.

Dropped Assumptions

- **Assumption A2 (Molecular Clock):** All terminal nodes are sampled at the same evolutionary time $\tau = 0$, implying strictly equal path lengths to the root.
- **Constant Block Model:** Cross-clan similarities collapse to a single flat value $S_{ij} = \beta_0$.
- **2-Plateau Fiedler Vector:** The population Fiedler vector $v^{(2)}$ is perfectly piecewise constant (c_A on clan A, $-c_B$ on clan B).

New Assumptions

- **Bounded Block Model:** Cross-clan similarities fluctuate but are strictly bounded within a narrow window: $S_{\text{out}}^{\min} \leq S_{ij} \leq S_{\text{out}}^{\max}$.
- **Condition C1 (Separation Margin) Before vs. Now:**
 - **Old C1:** Under the molecular clock, cross-clan similarities collapse to β_0 . The margin ρ simplified to $S_{\text{in}}^{\min} - \beta_0$. Condition C1 ($\rho > \eta S_{\text{out}}^{\max}$) was exactly $S_{\text{in}}^{\min} > (1 + \eta)\beta_0$.
 - **New C1:** The structural margin ρ includes a penalty for cross-clan variation. The new uncollapsed condition must absorb this variation: $(S_{\text{in}}^{\min} - S_{\text{out}}^{\max}) - (S_{\text{out}}^{\max} - S_{\text{out}}^{\min}) > \eta S_{\text{out}}^{\max}$.
- **Bounded Intra-Clan Fluctuation:** The departure from the molecular clock is mild enough that the Fiedler vector's intra-clan fluctuation $\epsilon = \max_{i,j \in \text{clan}} |v_i^{(2)} - v_j^{(2)}|$ is bounded by $\mathcal{O}(1/\sqrt{m})$, and in fact stays strictly below the smaller plateau, $\epsilon \leq c_B/2$, so that every clan-B entry retains the sign of its plateau.

2 Modified Proof Outline (Annotated Changes)

The proof compares the entry-wise error $|\hat{v}_i^{(2)} - v_i^{(2)}|$ against the entry-wise margin $|v_i^{(2)}|$. It runs in four stages.

Stage 1 (population geometry and noise budget). The imbalance η sets both sides of the condition. It fixes the margin and the spectral gap through the ~~piecewise-constant~~ **nearly piecewise-constant** Fiedler vector and the gap bound ~~kept positive by Condition C1~~ **kept positive by the New Condition C1**. On the other side, matrix Bernstein applied to the sampling mask bounds the noise.

Stage 2 (entry-wise reduction). **The mathematical mechanism of Stage 2 remains unchanged.** Since the recovery event is in ℓ_∞ while Davis-Kahan controls only ℓ_2 , we use a Neumann resolvent expansion. This identically trades the operator norm for a single scalar per coordinate, $(E_L v^{(2)})_i$. **While the algebra holds exactly as before, the population Fiedler vector $v^{(2)}$ inside this expansion is no longer perfectly piecewise constant.**

Stage 3 (per-node concentration). The surviving scalar $(E_L v^{(2)})_i$ is a sum of independent, mean-zero increments, each carrying the factor $v_i^{(2)} - v_j^{(2)}$.

Every within-clan increment therefore vanishes exactly — this is a cancellation, not a bound, and it rests on the clock making $v^{(2)}$ exactly piecewise constant. Only cross-clan pairs survive, and Bernstein’s inequality applies node by node.

Variance Before vs. Now:

- **Old Variance (σ_{old}^2):** ~~Because $v^{(2)}$ was exactly flat on each clan, $v_i^{(2)} - v_j^{(2)} = 0$ for all same-clan pairs. Intra-clan variance was exactly 0. The total per-node variance was entirely from cross-clan pairs: $\sigma_{old}^2 = \mathcal{O}(1/p)$.~~
- **New Variance (σ_{new}^2):** Because $v^{(2)}$ fluctuates, within-clan increments survive and inject $\mathcal{O}(m)$ new terms per node. Bounding the intra-clan fluctuation by ϵ , the new variance is:

$$\sigma_{new}^2 = \underbrace{\mathcal{O}\left(\frac{1}{p}\right)}_{\text{cross-clan}} + \underbrace{\mathcal{O}\left(\frac{m\epsilon^2}{p}\right)}_{\text{intra-clan}}$$

Bernstein’s inequality now applies to this much larger combined variance.

Stage 4 (synthesis). The per-node condition is tightest on clan B. Squaring the Bernstein deviation bound $\approx \sqrt{\sigma_{new}^2 \log m}$ against the spectral gap margin $\Delta\lambda_{c_B}$ yields the sampling rate.

~~The factor η cancels and solving for p gives exactly $p = \Theta\left(\frac{\log m}{m}\right)$.~~

Substituting the updated variance σ_{new}^2 and solving for p yields a degraded rate that explicitly depends on the structural deviation ϵ :

$$p = \Omega\left(\frac{\log m}{m} + \epsilon^2 \log m\right)$$

If the tree is near-ultrametric ($\epsilon \leq \mathcal{O}(1/\sqrt{m})$), the $\mathcal{O}(\frac{\log m}{m})$ scaling survives. The regime is bounded above by $\epsilon \asymp c_B$: there a clan-B entry of the *population* Fiedler vector crosses zero, the sign-partition is lost before any sub-sampling, and no rate p restores it.

A Revised Proof Outline (Clean Version for Paper)

The proof of Theorem 2 compares two quantities evaluated at the same node: the entry-wise error $|\hat{v}_i^{(2)} - v_i^{(2)}|$ and the entry-wise margin $|v_i^{(2)}|$. Under the relaxed bounded block model, the population Fiedler vector is no longer exactly piecewise constant. Let $\epsilon = \max_{i,j \in \text{clan}} |v_i^{(2)} - v_j^{(2)}|$ denote the maximum intra-clan fluctuation, assumed to satisfy $\epsilon \leq c_B/2$. The argument runs in four stages:

Stage 1 (population geometry and noise budget). The tree imbalance η fixes both the margin and the spectral gap $\Delta\lambda$. The uncollapsed structural margin $\rho = (S_{\text{in}}^{\min} - S_{\text{out}}^{\max}) - (S_{\text{out}}^{\max} - S_{\text{out}}^{\min})$, kept strictly positive by the generalized Condition C1, ensures $\Delta\lambda > 0$. Concurrently, a matrix Bernstein bound applied to the sampling mask restricts the spectral norm of the noise, $\|E_L\|_2 \leq \tilde{C} \sqrt{\frac{m \log m}{p}}$.

Stage 2 (entry-wise reduction). Since the recovery condition $\|\hat{v}^{(2)} - v^{(2)}\|_\infty < \min_i |v_i^{(2)}|$ is an ℓ_∞ event, standard Davis-Kahan ℓ_2 bounds are too loose. Provided $\|E_L\|_2 < \Delta\lambda$, a Neumann resolvent expansion isolates the entry-wise error to first order. This trades the operator norm for a single scalar per coordinate, reducing the target condition to $|(E_L v^{(2)})_i| \lesssim \Delta\lambda |v_i^{(2)}|$ holding at every node i .

Stage 3 (per-node concentration). Target: $|(E_L v^{(2)})_i|$, at every node.

1. *Expand.* With $L = D - S$, $(E_L v^{(2)})_i = \sum_{j \neq i} X_{ij}$, where $X_{ij} = \left(\frac{B_{ij}}{p} - 1\right) S_{ij}(v_i^{(2)} - v_j^{(2)})$ and $B_{ij} \sim \text{Bernoulli}(p)$: independent, mean-zero, and weighted by the Fiedler gap that the clock used to annihilate.
2. *Sum the variances.* $\text{Var}(X_{ij}) = \frac{1-p}{p} S_{ij}^2 (v_i^{(2)} - v_j^{(2)})^2 \asymp (v_i^{(2)} - v_j^{(2)})^2 / p$; cross-clan pairs carry $\Theta(1/m)$ each, same-clan pairs at most ϵ^2 each, and there are $\Theta(m)$ of both:

$$\sigma^2 \approx \mathcal{O}\left(\frac{1}{p}\right) + \mathcal{O}\left(\frac{m\epsilon^2}{p}\right), \quad (1)$$

the two terms balancing at $\epsilon = \Theta(1/\sqrt{m})$.

3. *Concentrate.* $|X_{ij}| \leq b = \mathcal{O}(1/(p\sqrt{m}))$, so Bernstein with a union bound over the m nodes gives, with probability $1 - m^{-c}$,

$$\left|(E_L v^{(2)})_i\right| \lesssim \sqrt{\sigma^2 \log m} + b \log m \asymp \sqrt{\sigma^2 \log m} \quad \text{for } p = \Omega(\log m / m). \quad (2)$$

Stage 4 (synthesis). The per-node condition is tightest on the larger clan B , where the entry-wise margin is smallest. That margin is now the worst entry rather than a plateau value: $v_{\min} := \min_{i \in B} |v_i^{(2)}| \geq c_B - \epsilon \geq c_B/2 = \Theta(1/\sqrt{m})$, the last step by the bounded-fluctuation assumption. Recovery requires the Bernstein deviation to stay below the gap-weighted margin, $\sqrt{\sigma^2 \log m} \lesssim \Delta\lambda v_{\min}$. Both sides carry an explicit m -scaling. On the right, L is the unnormalized Laplacian of a similarity matrix with $\Theta(1)$ entries, so its degrees — and hence the gap separating $v^{(2)}$ — grow with the block size: $\Delta\lambda = \Theta(\rho m)$. On the left, $\sigma^2 \approx \frac{1}{p}(1 + m\epsilon^2)$ from Stage 3. Squaring the condition gives

$$\frac{\log m}{p} (1 + m\epsilon^2) \lesssim \Delta\lambda^2 v_{\min}^2 = \Theta(\rho^2 m), \quad (3)$$

and solving for the sampling rate,

$$p = \Omega\left(\frac{\log m}{\rho^2} \left(\frac{1}{m} + \epsilon^2\right)\right) = \Omega\left(\frac{\log m}{m} + \epsilon^2 \log m\right), \quad (4)$$

the last equality at a fixed structural margin $\rho = \Theta(1)$. The two terms are the two variance sources of Stage 3, and the deviation ϵ decides which dominates: for near-ultrametric topologies with $\epsilon \leq \mathcal{O}(1/\sqrt{m})$

the cross-clan term leads and the $\mathcal{O}(\frac{\log m}{m})$ scaling is preserved; as ϵ grows the intra-clan term takes over and drives p toward $\Theta(1)$. The construction is valid up to $\epsilon \asymp c_B$, where $v_{\min}$ vanishes: there the population Fiedler vector itself stops separating the clans by sign, so the boundary is one of identifiability rather than of sampling rate.